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Chapter 7 - Problems and Solutions

Problem 7.1: A horizontal pipeline has a diameter of 0.1 m. Water is pumped through this pipeline with a mean velocity V=1.35 m/sec. Kinematic viscosity index of water $\nu=10^{-7}$ m^2/sec. Roughness of the boundary a=1.5 10^{-4} m.

a) What difference in head can be read from two manometers that are connected to the pipe and of which the distance between the points of connection equals to 100 m.

b) Same question for a liquid with a viscosity index. $\nu=10^{-5}$ m^2/s

c) Same question for a liquid with a viscosity index. $\nu=10^{-4}$ m^2/s

Solution:

a) Reynolds number $\mathrm{Re} = \dfrac{VD}{\nu} = \dfrac{1.35 \times 0.1}{10^{-7}} = 1.35 \times 10^{6} > 2320$ flow is turbulent.

Assume hydraulic rough flow.

$$C = 18 \log \frac{6R}{a} = 18 \log \frac{6 \times \frac{0.1}{4}}{1.5 \times 10^{-4}} = 54\,\mathrm{m}^{1/2}/\mathrm{s}$$

$V = C\sqrt{RI}$ or $I = \dfrac{V^2}{C^2 R}$ or $\Delta h_L = \dfrac{L V^2}{C^2 R}$

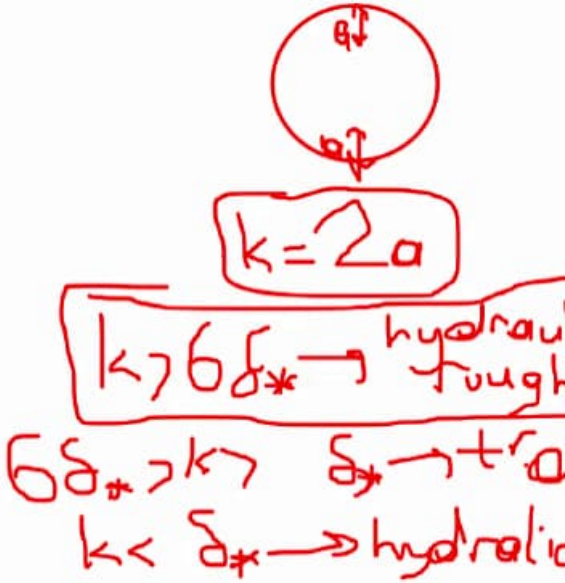
$$\Delta h_L = \frac{100 \times 1.35^2}{54^2 \times \frac{0.1}{4}} = 2.5\,\text{m}$$

Check the assumption;

$$u_* = \sqrt{g R I} = \sqrt{10 \times \frac{0.1}{4} \times \frac{2.5}{100}} = \frac{1}{12.6} = 0.079\ \text{m/s}$$

$$\delta_* = 11.6 \frac{\nu}{u_*} = 11.6 \times \frac{10^{-7}}{1/12.6} = 1.46 \times 10^{-5}\ \text{m}$$

$k = 3 \times 10^{-4}\ \text{m} > 6\delta_* = 8.76 \times 10^{-5}\ \text{m}$ so the assumption is correct.



b) For $\nu=10^{-5}$ m/s^2

$$\mathrm{Re}=\frac{VD}{\nu}=\frac{1.35\times 0.1}{10^{-5}}=1.35\times 10^{4}>2320 \text{ turbulent flow}$$

Probably in transition zone between hydraulic rough and smooth. We need hydraulic gradient to calculate the thickness of viscous sub-layer. However, we have to find head loss so the

iteration is necessary.

Assume $\Delta h_L = 3$ m

$$u_* = \sqrt{10 \times \frac{0.1}{4} \times \frac{3}{100}} = 0.086 \text{ m/s}$$

$$\delta_* = 11.6 \times \frac{10^{-5}}{0.086} = 1.32 \times 10^{-3} \text{ m}$$

$$C = 18\log\frac{6R}{a + \frac{\delta_*}{7}} = 18\log\frac{6 \times \frac{0.1}{4}}{1.5 \times 10^{-4} + \frac{1.32 \times 10^{-3}}{7}} = 47.5 \text{ m}^{1/2}/\text{s}$$

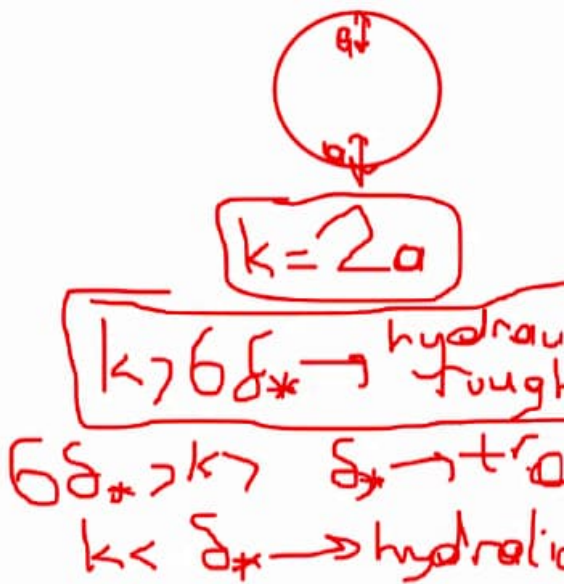
Check:

$$\Delta h_L = \frac{L\,\bar{V}^2}{C^2\,R} = \frac{100 \times 1.35^2}{47.5^2 \times \frac{0.1}{4}} = 3.22 \neq 3 \text{ m}$$

so new head loss is $\Delta h_L = 3.22$ m

$$u_* = \sqrt{10 \times \frac{0.1}{4} \times \frac{3.22}{100}} = 0.09 \text{ m/s}$$

$$\delta_* = 11.6 \times \frac{10^{-5}}{0.09} = 1.29 \times 10^{-3} \text{ m}$$



$$C = 18\log\frac{6\times\frac{0.1}{4}}{1.5\times10^{-4}+\frac{1.29\times10^{-3}}{7}} = 47.5\,\mathrm{m}^{1/2}/\mathrm{s}$$

Check:

$$\Delta h_L = \frac{100\times1.35^2}{47.5^2\times\frac{0.1}{4}} = 3.22\ \mathrm{m}$$

$k = 2a$

$k > 6\delta^* \rightarrow$ hydrau
tough

$6\delta^* > k > \delta^* \rightarrow$ tra

This gives the result because assumption is correct (there is no difference).

c) For $v=10^{-4}$ m^2/s

$$\mathrm{Re}=\frac{1.35\times 0.1}{10^{-4}}=1350<2320$$ Laminar flow

$u_* = \sqrt{g \cdot R \cdot I}$

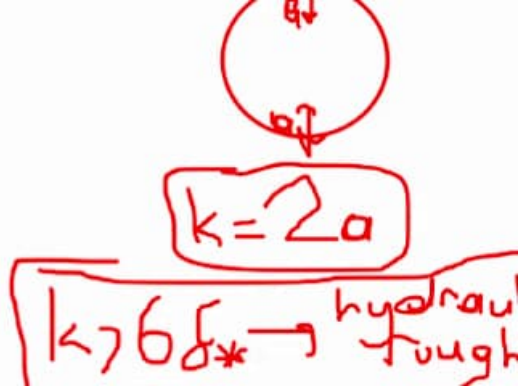
$\delta_* = 11{,}6 \cdot \frac{v}{u_*}$

$\frac{\delta R}{a+\frac{\delta_*}{7}}$

From Hagen Poiseuille relation, average flow

$$V = g\frac{I}{\nu}\frac{D^2}{32} = \frac{10\times\frac{\Delta h_L}{100}}{10^{-4}}\times\frac{0.1^2}{32} = 1.35\,m/s \quad \text{and}$$

Δh_L=4.32 m.



Problem 7.2: In a welded steel oil line (diameter 0.6m), pumps are spaced 10 km apart. The pumps are working a difference in pressure head of 25 m oil column. Calculate the discharge. Density of the oil is $\rho=800$ kg/m^3, kinematic viscosity is $\nu=10^{-4}$ m^2/s, boundary roughness is $a=5\times10^{-5}$ m. In the course of time, the roughness of the pipe (expressed in terms of roughness length) will grow 3×10^2 times the original value. Will the discharge be influenced by this? If so, how much (in %) will the reduction (approximate) be?

Solution:

Pipe diameter $D=0.6$ m

Pipe length $L=10^4$ m

Pressure Head Difference $\frac{\Delta p}{\gamma}=25$ m

so friction loss $\Delta h_L = \frac{\Delta p}{\gamma} = 25$ m (Pipe diameter is constant).

Hydraulic radius $R = \frac{D}{4} = \frac{0.6}{4} = 0.15$ m

Shear velocity $u_* = \sqrt{g R I} = \sqrt{10 \times 0.15 \times 25 \times 10^{-4}} = 6 \times 10^{-2}$ m/s

The thickness of viscous sub-layer $\delta_* = 11.6 \frac{\nu}{u_*} = 11.6 \times \frac{10^{-4}}{6 \times 10^{-2}} = 2 \times 10^{-2}$ m

$k=10^{-4}\ \text{m} < \delta_*=2\times10^{-2}\ \text{m}$ Hydraulic smooth

$$C=18\log\frac{6R}{a+\frac{\delta_*}{7}}=18\log\frac{6R}{\frac{\delta_*}{7}}=18\log\frac{6\times0.15}{3\times10^{-3}}=44.6\ \text{m}^{1/2}/\text{s}$$

$$V=C\sqrt{RI}=44.6\sqrt{0.15\times25\times10^{-4}}=0.86\ \text{m/s}$$

$$Q=VA=V\times\frac{\pi D^2}{4}=0.86\times\frac{\pi\times0.6^2}{4}=0.24\ \text{m}^3/\text{s}$$

Now again we will do the calculation for new friction case.

$a_{new}=300\times=1.5\times10^{-2}$ m,

$6\delta_*=1.2\times10^{-1}$ m

$k=3\times10^{-2}$ m,

$\delta_* < k < 6\delta_*$ transition zone

$$C = 18\log\frac{6\times0.15}{1.8\times10^{-2}} = 30.6\ \mathrm{m}^{1/2}/\mathrm{s}$$

for the new case (Q_2)

$$\frac{Q_2}{Q_1} = \frac{C_2\sqrt{RI}\,A_2}{C_1\sqrt{RI}\,A_1}$$

$$\frac{Q_2}{Q_1} = \frac{C_2}{C_1}$$

$$\frac{Q_2}{Q_1} = \frac{30.6}{44.6} = 0.69$$

$$1 - \frac{Q_2}{Q_1} = 1 - 0.69 = 0.31$$

discharge is changed and reduced 31%.

Problem 7.3: Find the discharge of the pipe system shown in Figure 7.3 and if the diameter of the system is converted into ϕ150 mm find the length of the pipe of the system.

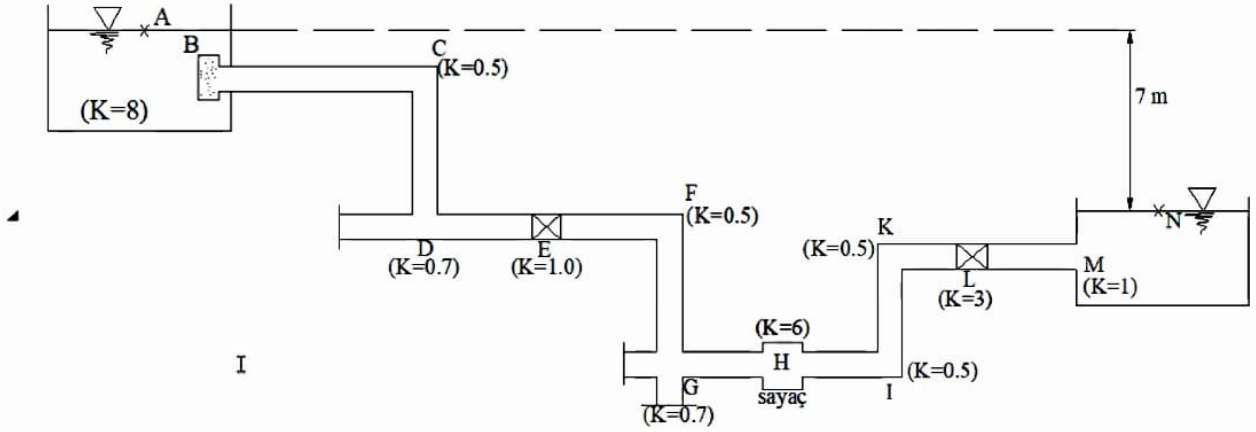


Figure 7.3

Between B-G (1)

L=46 m λ=0.025 D=300 mm

Between G-M (2)

L=30 m λ=0.020 D=150 mm

Solution:

Energy equation between A-N

$$\frac{V_A^2}{2g}+\frac{P_A}{2g}+z_A=\frac{V_N^2}{2g}+\frac{P_N}{2g}+z_N+\sum\Delta h$$

$$\sum\Delta h=\sum K_1\frac{V_1^2}{2g}+\lambda_1\frac{L_1}{D_1}\frac{V_1^2}{2g}+\sum K_2\frac{V_2^2}{2g}+\lambda_2\frac{L_2}{D_2}\frac{V_2^2}{2g}$$

$$\sum \Delta h = \frac{V_1^2}{2g}\left(\sum K_1 + \lambda_1 \frac{L_1}{D_1}\right) + \frac{V_2^2}{2g}\left(\sum K_2 + \lambda_2 \frac{L_2}{D_2}\right)$$

Continuity equation

$Q = A_1V_1 = A_2V_2$

$$\frac{\pi \times 0.3^2}{4} \times V_1 = \frac{\pi \times 0.15^2}{4} \times V_2$$

$$V_1 = \frac{1}{4} V_2 \quad \text{veya} \quad V_1^2 = \frac{1}{16} V_2^2$$

So

$$\sum \Delta h = \frac{V_2^2}{32g}\left(\sum K_1 + \lambda_1 \frac{L_1}{D_1}\right) + \frac{V_2^2}{2g}\left(\sum K_2 + \lambda_2 \frac{L_2}{D_2}\right)$$

$$\sum \Delta h = \frac{V_2^2}{2g}\left[\frac{1}{16}\left(\sum K_1 + \lambda_1 \frac{L_1}{D_1}\right) + \left(\sum K_2 + \lambda_2 \frac{L_2}{D_2}\right)\right]$$

$$7 = \frac{V_2^2}{19.62}\left[\frac{1}{1.0}\left(8+0.5+0.7+1+0.5+0.025\frac{46}{0.3}\right)+\left(0.7+6.0+0.5+0.5+3.0+1.0+0.02\frac{30}{0.15}\right)\right]$$

$$7 = 16.6 \times \frac{V_2^2}{19.62}$$

V_2=2.88 m/s

$$Q=\frac{\pi\times 0.15^2}{4}\times 2.88=0.05\,m^3/s$$

If the system is converted into an equivalent pipe system

$$16.6\times\frac{V_2^2}{19.62}=0.020\times\frac{L}{0.15}\times\frac{V_2^2}{2g}$$

$E_{equivalent}$=125 m